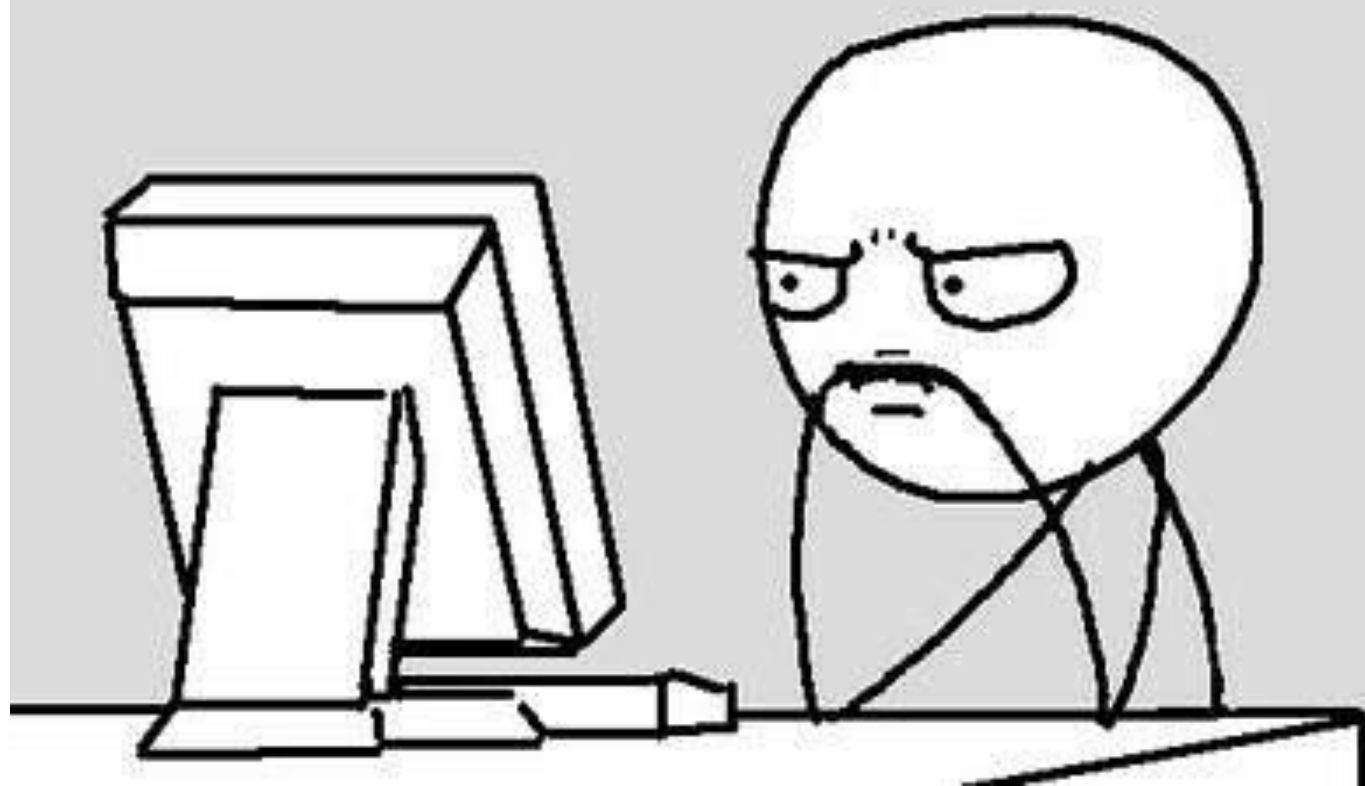


LET'S WAIT



#LifeKoKaroLift

Data Science Certification Program



Course : Machine Learning
Lecture On : Hypothesis
Testing

Instructor : Shivam Garg

No
Validation

Validation

Problem Statement ✓

Inferential Statistics (Sample Study) (CLT)

Inference (hypothesis) } (hypothesis about population)

Hypothesis testing (perform the test)

Whether hypothesis is correct or not → Decision making

Hypothesis Testing: (validated)

Inferences about population derived with the help of inferential statistics.

H_0 : Null hypothesis: { Status quo (Claim given)

H_1 : Alternate hypothesis: { Opposite to Status quo
(Theoretical definition)

{ Numericals (Symbol

{ =, ≥, ≤ }

↓
 H_0

{ ≠, >, < }

↓
 H_1

always ✓

↔ Claim: fK will generate average ticket size of 2000?

T { $H_0 = \mu = 2000$ ←
 $H_1 = \mu \neq 2000$

S { $H_0 = \mu = 2000$
✓ $H_1 = \mu \neq 2000$

→ Claim: fK will not generate average ticket size of 2000?

T { $H_0 = \mu \neq 2000$
✓ $H_1 = \mu = 2000$

S { $H_0 = \mu = 2000$
✓ $H_1 = \mu \neq 2000$

Formulating Null and Alternate Hypothesis

Example-1: A restaurant owner installed a new automated drink machine. The machine is designed to dispense 530 mL of liquid on the medium size setting. The owner suspects that the machine may be dispensing too much in medium drinks. They decide to take a sample of 30 medium drinks to see if the average amount is significantly greater than 530 ml.

$$\left\{ \begin{array}{l} H_1 \rightarrow \mu > 530 \text{ ml} \\ H_0 \rightarrow \mu \leq 530 \text{ ml} \end{array} \right\} \text{formulat}^n \text{ of hypothesis}$$

$\rightarrow H_0: =, \geq, \leq$
$\rightarrow H_1: \neq, <, >$
$\rightarrow \text{Right tail test}$

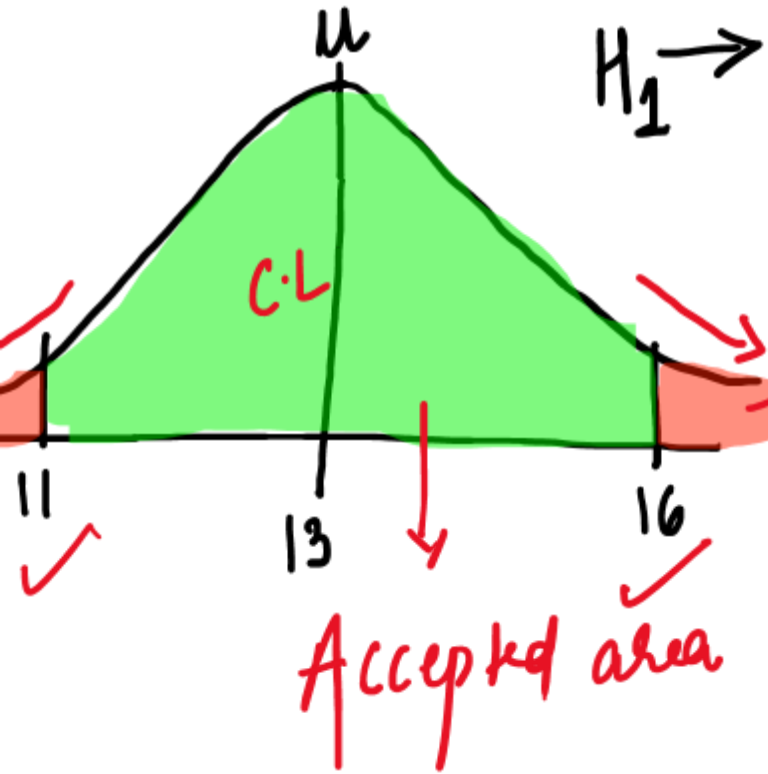
→ Hypothesis testing :- (Problem)

- ① formulation of hypothesis (what)
- ② Which test?
 - 2-Tail
 - Left (which)
 - Right
- ③ Methodology (CRM / p-value method) (how)
- ④ Decisioning

Type of test

Two tail Test ✓ $H_0 \rightarrow =$

$H_1 \rightarrow \neq$



One Tail test

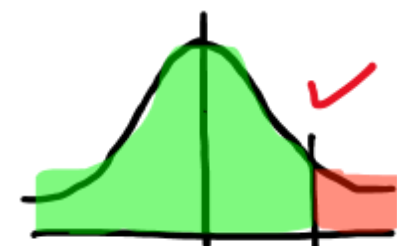
$\left\{ \begin{array}{l} H_0 \rightarrow \leq \\ H_1 \rightarrow > \end{array} \right.$

$\left\{ \begin{array}{l} H_0 \rightarrow \geq \\ H_1 \rightarrow < \end{array} \right.$

left tail test ✓



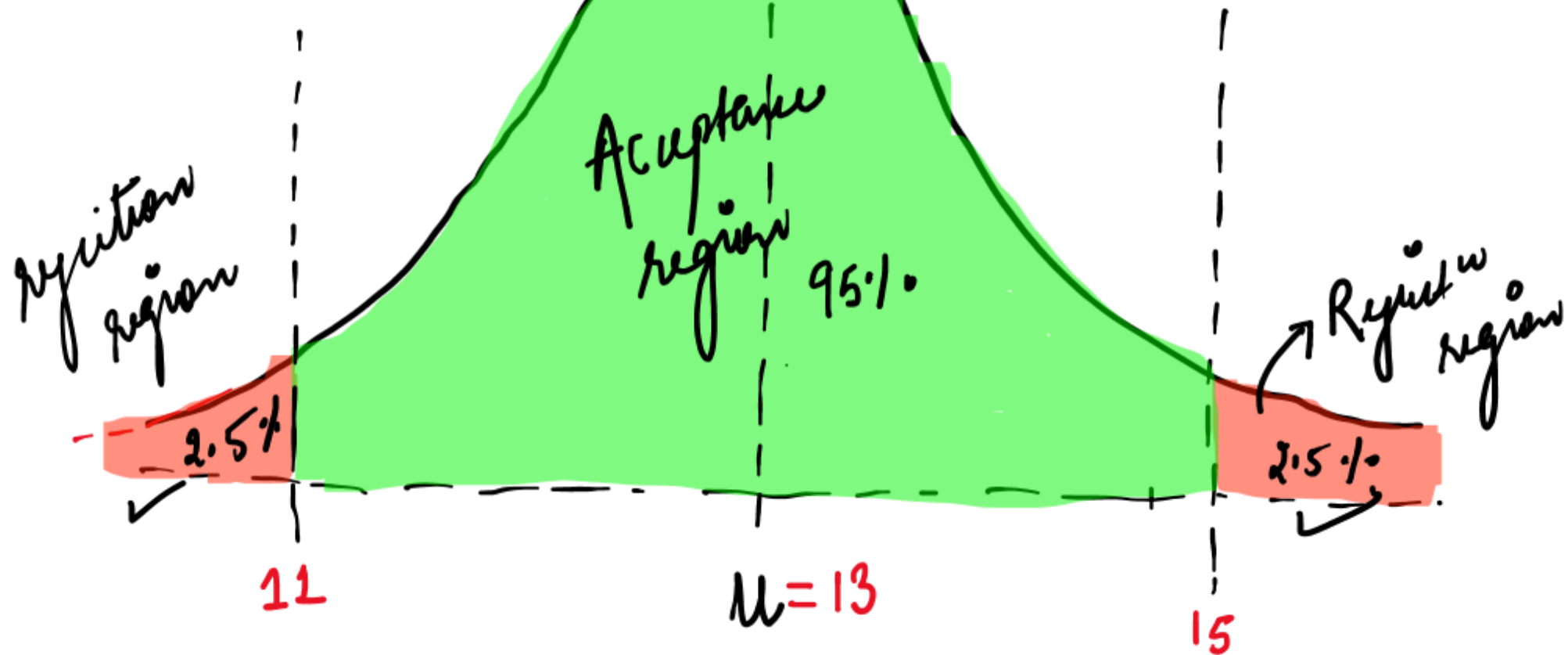
$\left\{ \begin{array}{l} \text{Right tail test} \end{array} \right.$



Confidence level = 95%
significance level = 5%
 α

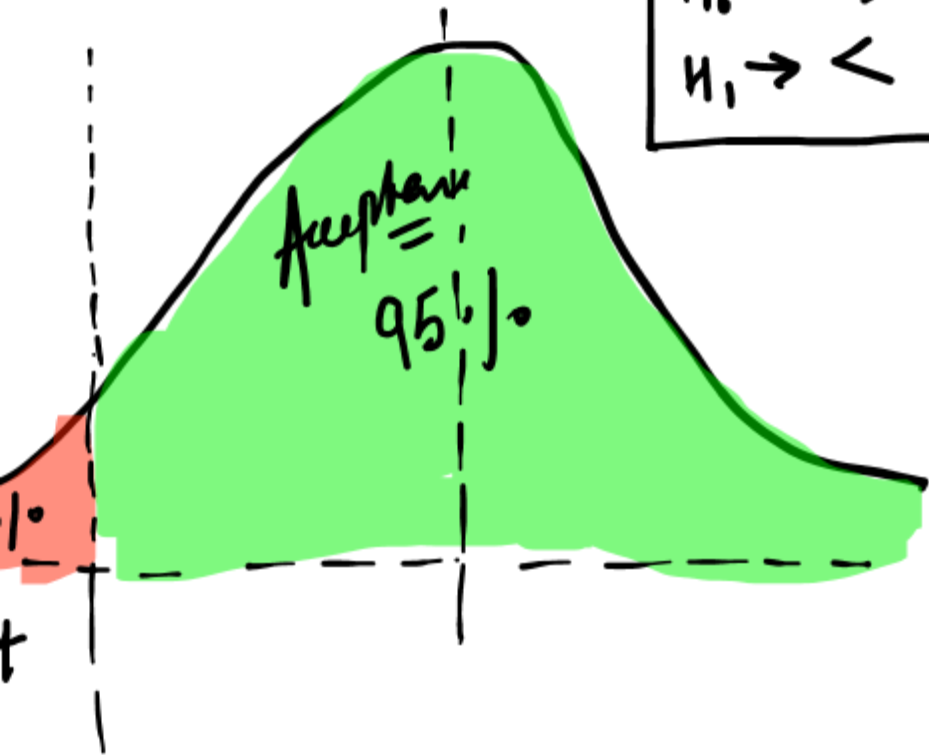
Two Tail Test $\rightarrow \left\{ \begin{array}{l} H_0 \rightarrow = \\ H_1 \rightarrow \neq \end{array} \right\}$

$\rightarrow$ 2 rejection area



Left Tail C.F. = 95%

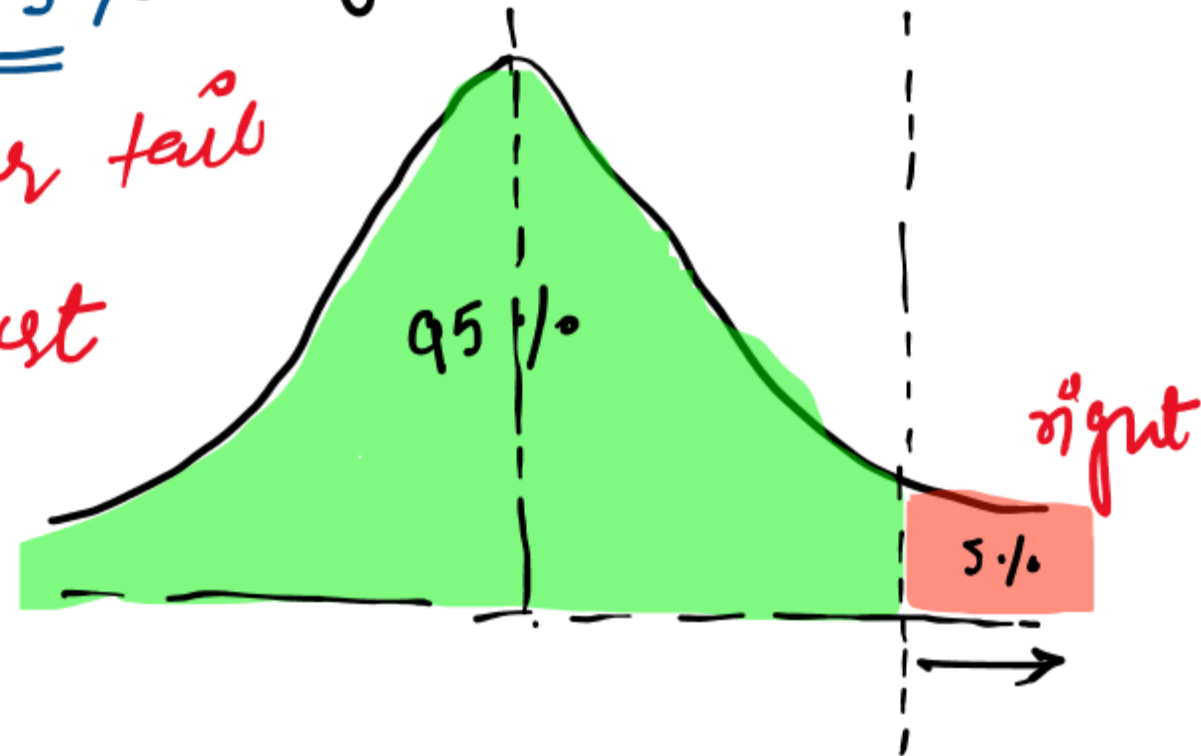
$$\begin{array}{l} H_0 \rightarrow \geq \\ H_1 \rightarrow < \end{array}$$



Lower tail test

C.L. = 95% Right Tail
S.L. = 5%
$$\begin{array}{l} H_0 \rightarrow \leq \\ H_1 \rightarrow > \end{array}$$

Upper tail test



$H_1 \rightarrow \neq \rightarrow$ 2-tail test

$H_1 \rightarrow < \rightarrow$ Left tail test

$H_1 \rightarrow > \rightarrow$ Right tail test

Two methods

```
graph TD; A["Two methods"] --> B["Critical Value method"]; A --> C["p-value method"];
```

Critical Value
method

p-value
method

→ Critical Value method :-

① formulation of hypothesis

② which test?

③ Calculate lower critical value and/or upper critical value
(L.C.V) (U.C.V) / (H.C.V)

2-tail test → L.C.V and U.C.V

Left tail test → L.C.V

Right tail test → U.C.V

$$L.C.V = \mu - Z \times \frac{\sigma}{\sqrt{n}}$$

$$U.C.V = \mu + Z \times \frac{\sigma}{\sqrt{n}}$$

A simple random sample of 50 adults women is obtained, and each person's red blood cell count (in cells per microliter) is measured. The sample mean is 4.63. The population standard deviation for red blood cell counts is 0.54. Construct the 95% confidence interval estimate for the mean red blood cell counts of adults.

$$\begin{aligned} n &= 50 \checkmark & \rightarrow C.L. &= 95\% \text{ (Green Area)} \\ \bar{x} &= 4.63 \checkmark & \rightarrow \alpha &= 5\% \\ \sigma &= 0.54 \checkmark & \uparrow & \text{Significance level (100 - CL)} \end{aligned}$$

Q: Analyst claims that average red blood cell count is 4.63 cells/ μ
validate the claim using population parameter of (4.53) cells/ μ

Sol:-
① formulation of hypothesis : $H_0: \mu = 4.63 \text{ cells}/\mu\text{L}$ } ✓
 $H_1: \mu \neq 4.63 \text{ cells}/\mu\text{L}$

② Which test? $\rightarrow$ Two Tail test

$$\textcircled{3} \quad \text{L.C.V} = \mu - z \times \frac{\sigma}{\sqrt{n}} \quad \Bigg| \quad \text{H.C.V} = \mu + z \times \frac{\sigma}{\sqrt{n}}$$

→ Range Z-table

C.L ↑ 99% 2.58

C.L ↓ 90% 1.65

95% → 1.96

30 ± 3
 ↑ Answer
 (Sample)
 Error
 10%
 fix tolerance

30 ± Z × 3
tolerance

$\mu \pm Z \frac{\sigma}{\sqrt{n}}$
 Sampling error
 Error

$$\bar{X} = \mu \pm Z \times \frac{\sigma}{\sqrt{n}}$$

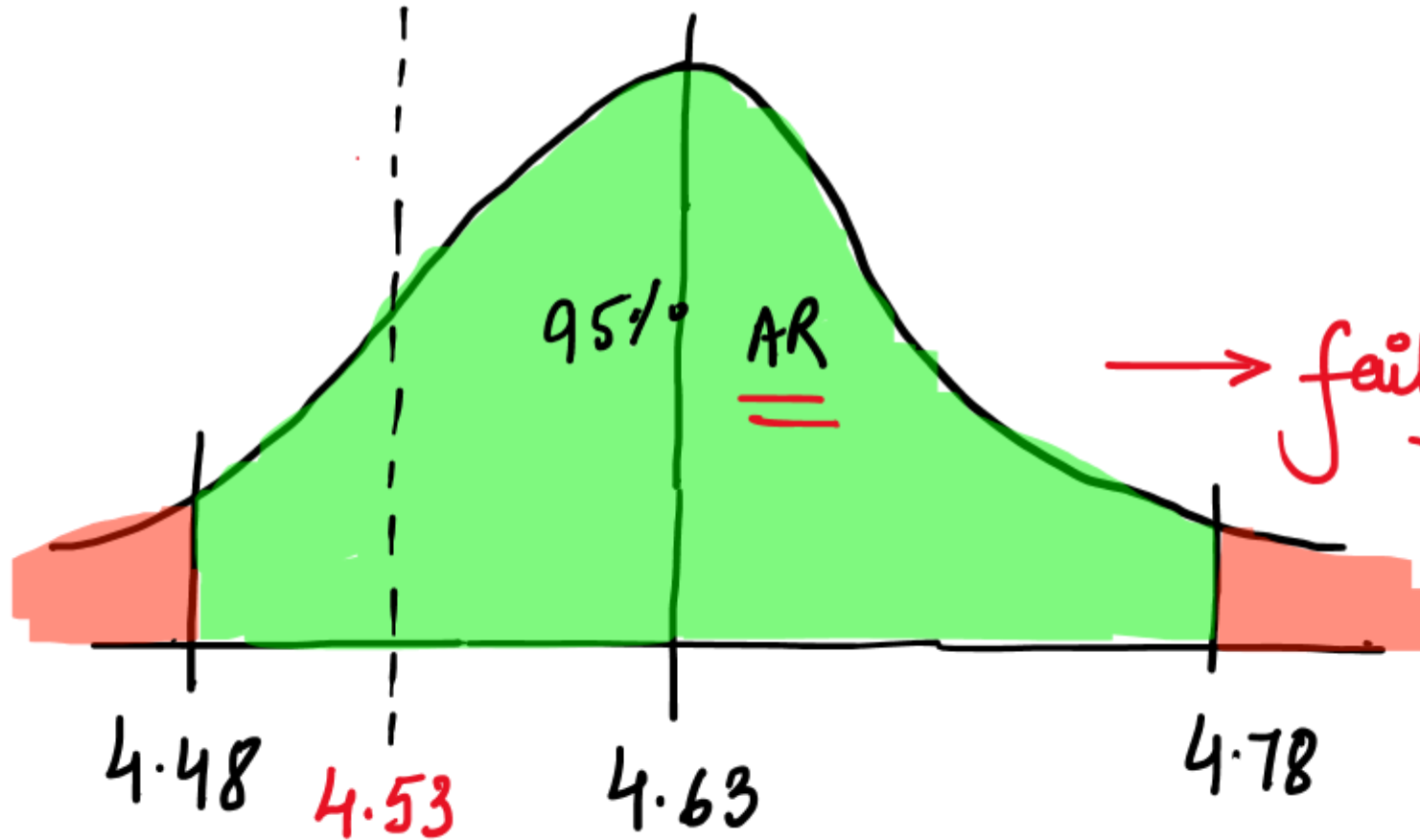
$$L.C.V = 4.63 - 1.96 \times \frac{0.54}{\sqrt{50}} \checkmark$$

$$= 4.48$$

$$H.C.V = 4.63 + 1.96 \times \frac{0.54}{\sqrt{50}}$$

$$= 4.78$$

④



$$4.63 \pm 0.15$$

→ fail to reject null hypothesis

→ p-value method:-

① formulation of hypothesis ✓

② which test ✓

③ Calculate z-stat →
$$Z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

④ calculate the p-value from z-stat using z-table.

⑤ Decisioning

$\bar{x}$: population parameter
 μ : sample mean
 σ : pop. std
 n : sample size

$$\rightarrow H_0 \rightarrow \mu = 4.63, \quad H_1 \rightarrow \mu \neq 4.63$$

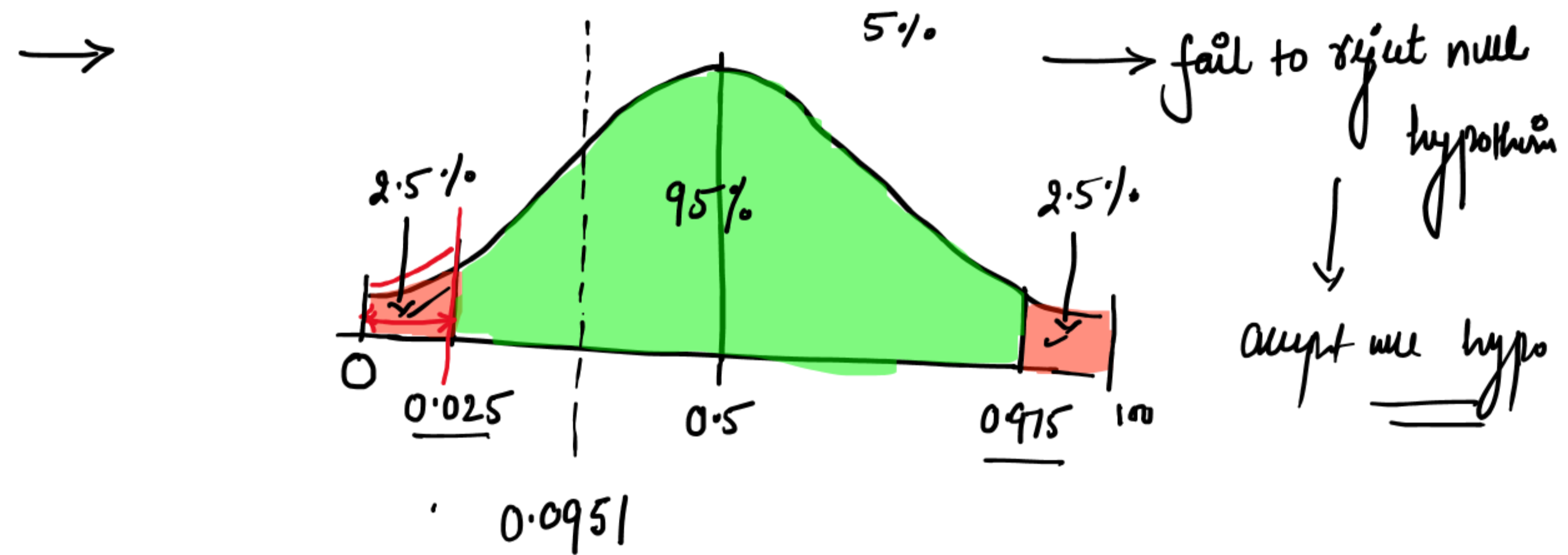
$\rightarrow$ Two-tailed test

$$\rightarrow Z\text{-Stat} = \frac{x - \mu}{\sigma / \sqrt{n}} = \frac{4.53 - 4.63}{0.54 / \sqrt{50}}$$

$$= \frac{-0.1 \times \sqrt{50}}{0.54} = - \frac{10 \times \sqrt{50}}{54} = -1.31$$

$$\left| \begin{array}{l} x = 4.53 \\ \mu = 4.63 \\ \sigma = 0.54 \\ n = 50 \end{array} \right.$$

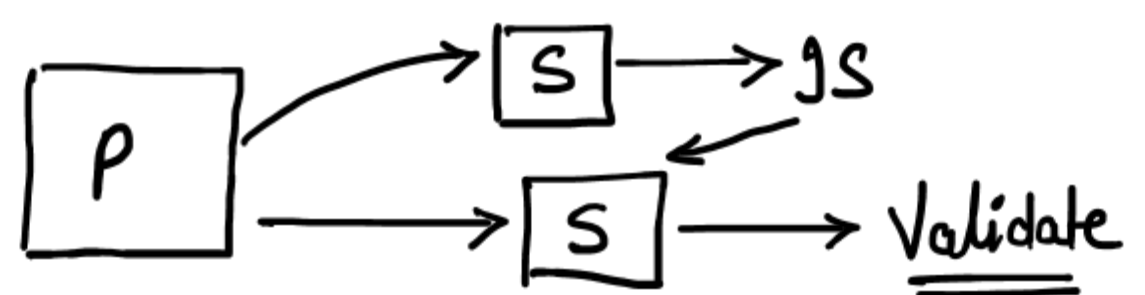
$$\rightarrow p\text{-value} (@ -1.31) = \underline{\underline{0.0951}}$$



* fail to reject null hypothesis } → Accept null hypothesis?

* Reject null hypothesis //

→ Type of Errors :-



Truth about the population
 ↓
H₀ true H_a true ←

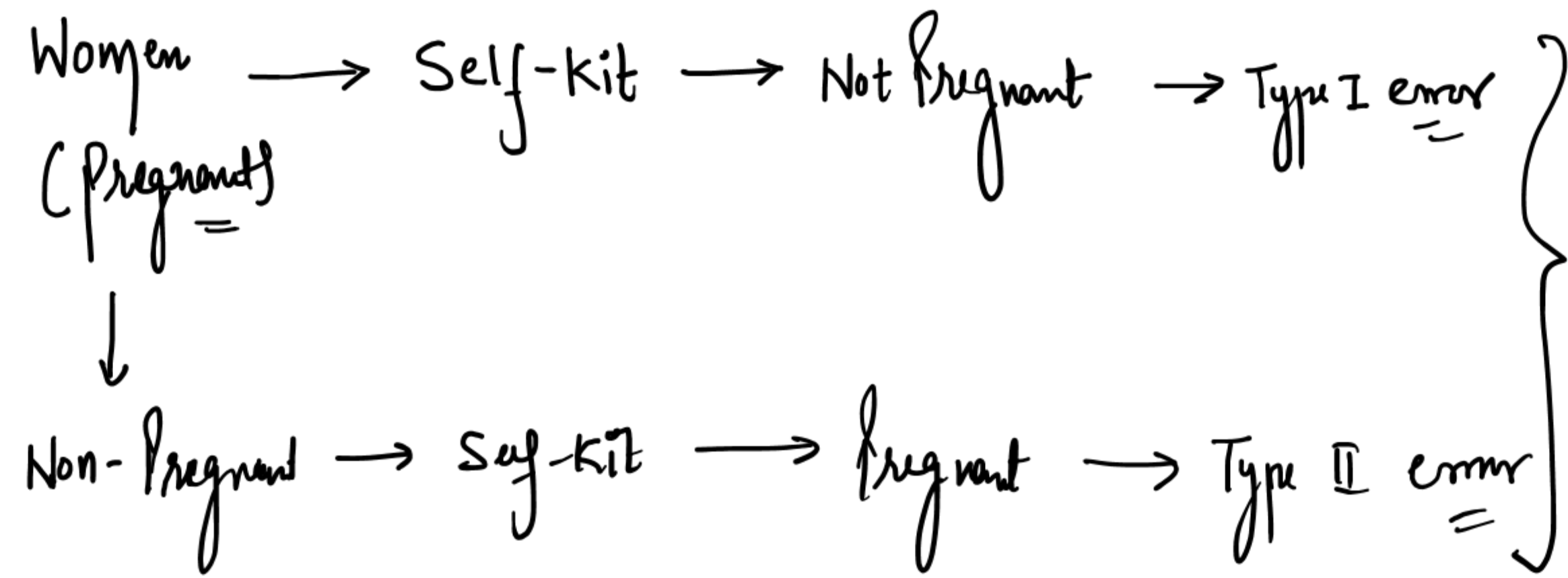
Decision based on sample
 → Reject H₀
 → Accept H₀

Type I error	Correct decision
Correct decision	Type II error

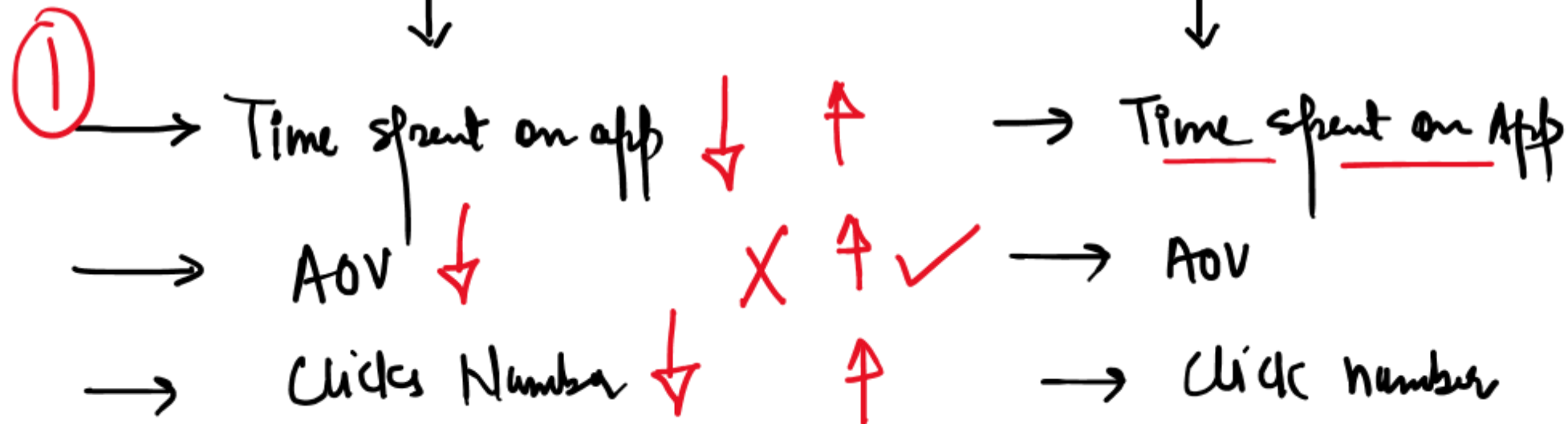
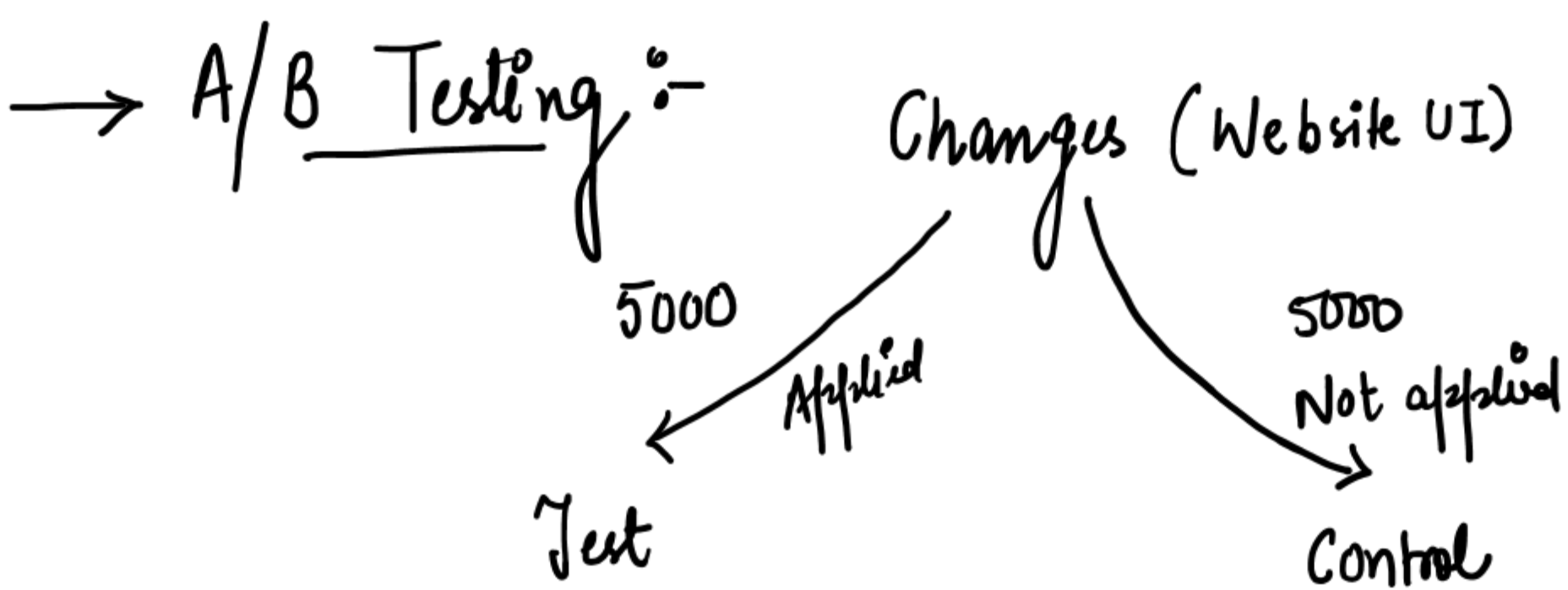
✓
4 B \$ → α → Type: Error

Claim ✓ → Rejected

Claim X → Accepted



ML $\rightarrow$ Logistic Reg



Interv → Def $\xrightarrow{HT}$ Not-def (Concern) Entire Amount ✓

→ Non-Def $\xrightarrow{HT}$ Def (Very less concern) Interest Am ✓

Telecom

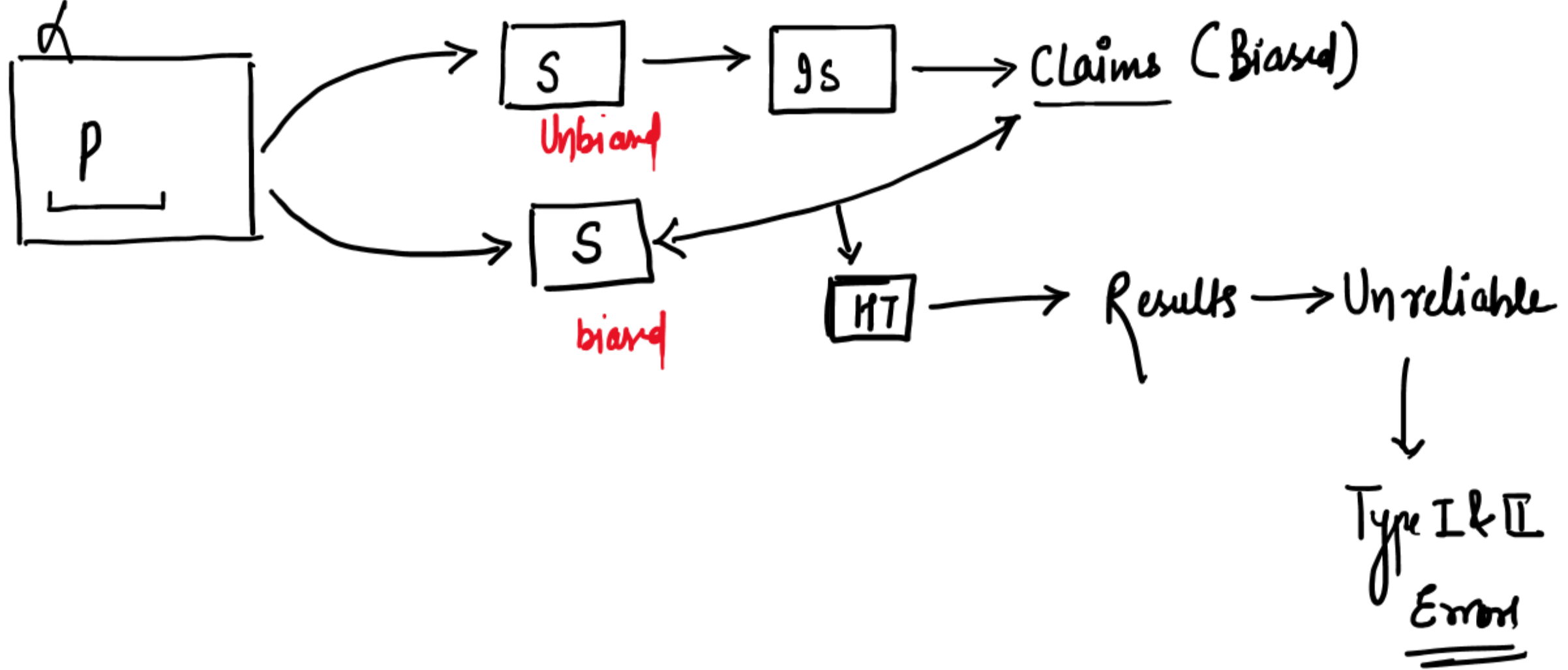
Churn $\xrightarrow{HT}$ Non-Churn (concern) ✓

Non-Churn $\xrightarrow{HT}$ Churn (Not a issue) of

H

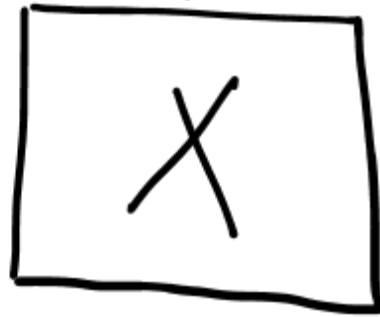
C → NC (concern)

NC → C (concern)

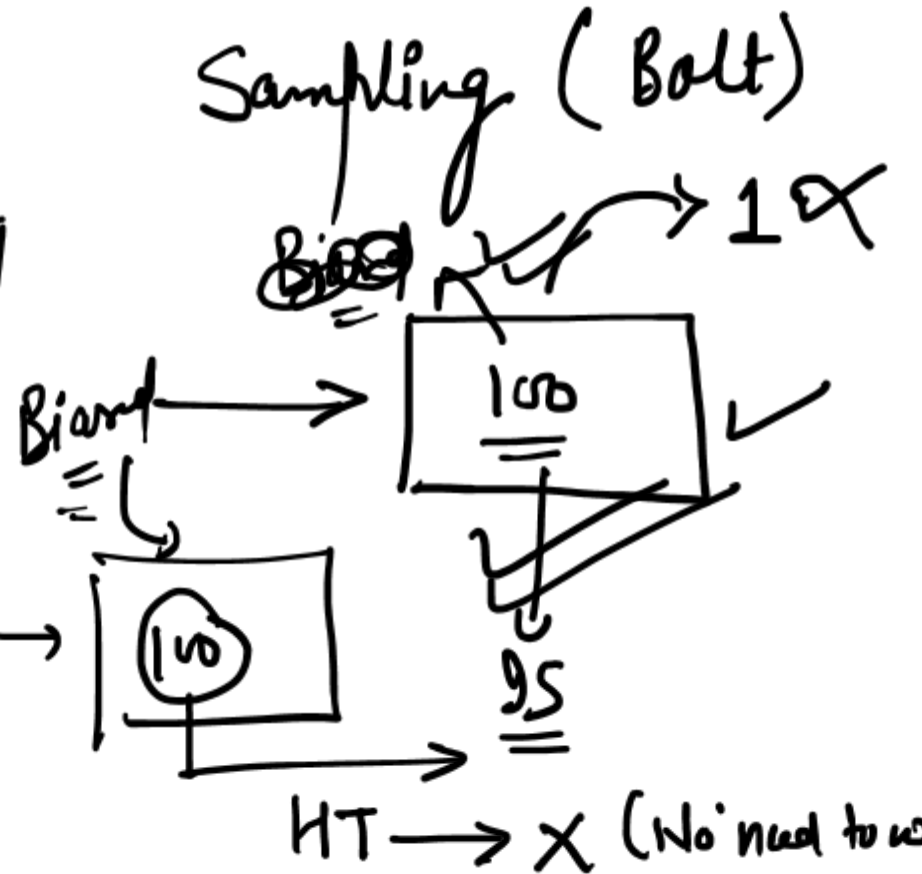


H_1 ✓
 α } =

$W = 5\text{kg} \rightarrow$ hypothesis is perfectly right



Block $\rightarrow$ claim $\rightarrow$ upto 10kg



$$\textcircled{1} \left. \begin{array}{l} \mu = 4.63 - H_0 \\ \mu \neq 4.63 - H_1 \end{array} \right\}$$

$\textcircled{2}$ 2-tail test

$$\textcircled{3} Z\text{-stat} = \frac{x - \mu}{\sigma / \sqrt{n}} = \frac{4.53 - 4.63}{0.54 / \sqrt{50}} = \underline{\underline{-1.31}}$$

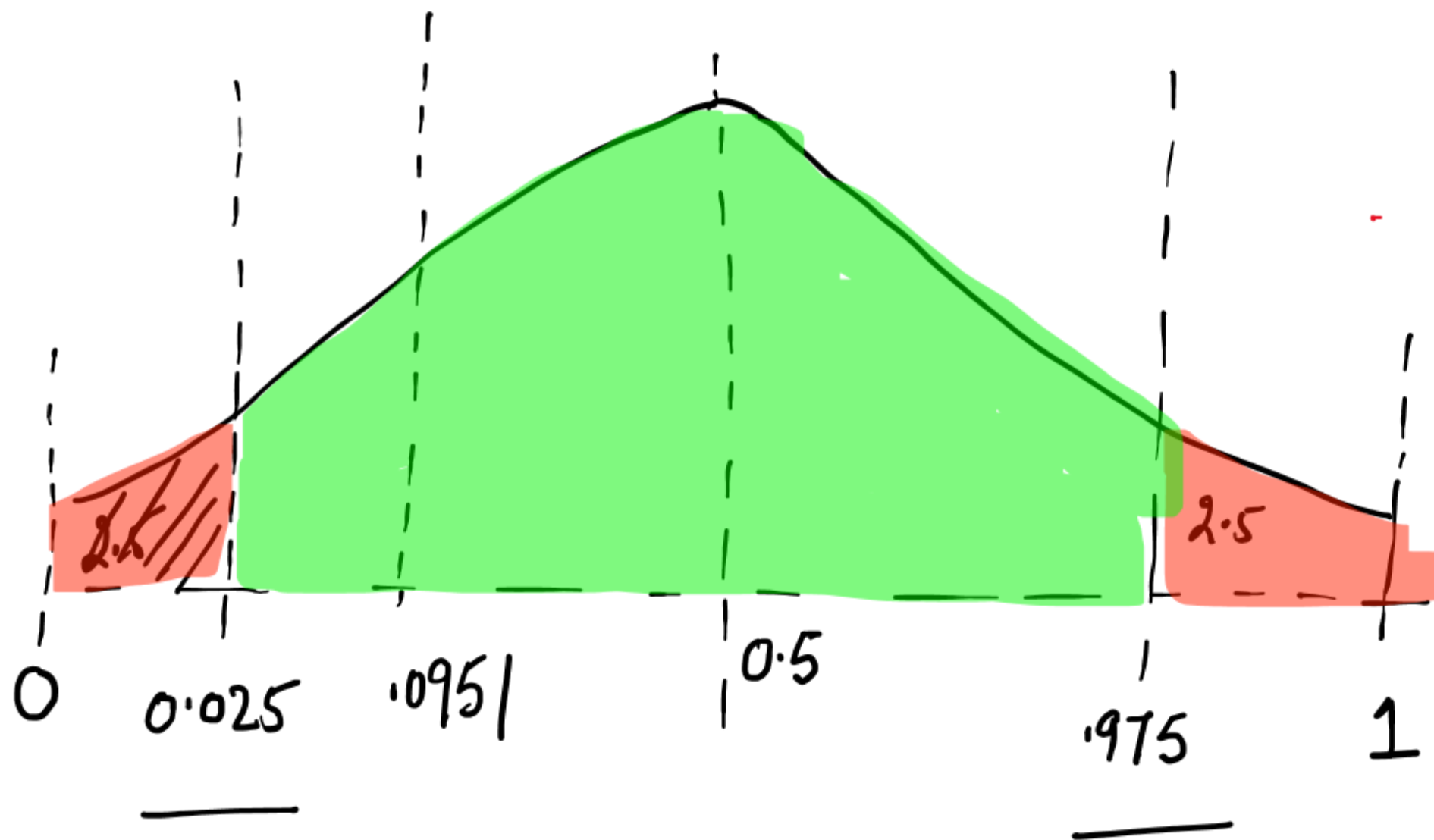
(Z-table)

$$\textcircled{4} p\text{-value} = 0.0951$$

(2-tailed)

$$\left. \begin{array}{l} x = 4.53 \\ \mu = 4.63 \\ \sigma = 0.54 \\ n = 50 \end{array} \right\}$$

CL = 95
 $\alpha = 5\%$



fail to reject the null hypo

$\alpha = 5\%$ p-value method

SUPERMARKET LOYALTY PROGRAM

Video: Speaker

- A supermarket plans to launch a loyalty program if:
Avg. spending per shopper > ₹120 per week
- A random sample of 80 shoppers enrolled in the pilot program spent an average of ₹130 in a week with a standard deviation of ₹40.

(a) formulation: $H_1: \mu > 120$

$H_0: \mu \leq 120$

(b) which test $\rightarrow$ Right tail test

Should the supermarket launch the loyalty program?

$$\begin{aligned} (c) Z_{Stat} &= \frac{\bar{x} - \mu}{\sigma / \sqrt{n}} = \frac{130 - 120}{40 / \sqrt{80}} = \frac{10}{40 / \sqrt{80}} = \frac{\sqrt{80}}{4} \\ &= \frac{4\sqrt{5}}{4} = \underline{\underline{2.23}} \end{aligned}$$

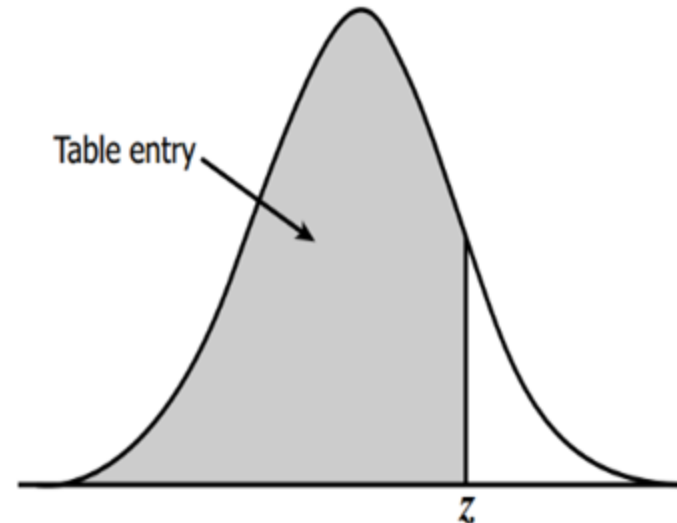
Z-table.

LOOKING UP Z-TABLE

z	.00	.01	.02	.03	.04	.05
0.0	.5000	.5040	.5080	.5120	.5160	.5199
0.1	.5398	.5438	.5478	.5517	.5557	.5596
0.2	.5793	.5832	.5871	.5910	.5948	.5987
0.3	.6179	.6217	.6255	.6293	.6331	.6368
0.4	.6554	.6591	.6628	.6664	.6700	.6736
0.5	.6915	.6950	.6985	.7019	.7054	.7088
0.6	.7257	.7291	.7324	.7357	.7389	.7422
0.7	.7580	.7611	.7642	.7673	.7704	.7734
0.8	.7881	.7910	.7939	.7967	.7995	.8023
0.9	.8159	.8186	.8212	.8238	.8264	.8289
1.0	.8413	.8438	.8461	.8485	.8508	.8531
1.1	.8643	.8665	.8686	.8708	.8729	.8749
1.2	.8849	.8869	.8888	.8907	.8925	.8944
1.3	.9032	.9049	.9066	.9082	.9099	.9115
1.4	.9192	.9207	.9222	.9236	.9251	.9265
1.5	.9332	.9345	.9357	.9370	.9382	.9394
1.6	.9452	.9463	.9474	.9484	.9495	.9505
1.7	.9554	.9564	.9573	.9582	.9591	.9599
1.8	.9641	.9649	.9656	.9664	.9671	.9678
1.9	.9713	.9719	.9726	.9732	.9738	.9744
2.0	.9772	.9778	.9783	.9788	.9793	.9798
2.1	.9821	.9826	.9830	.9834	.9838	.9842
2.2	.9861	.9864	.9868	.9871	.9875	.9878
2.3	.9893	.9896	.9898	.9901	.9904	.9906
2.4	.9918	.9920	.9922	.9925	.9927	.9929
2.5	.9938	.9940	.9941	.9943	.9945	.9946
2.6	.9953	.9955	.9956	.9957	.9959	.9960
2.7	.9965	.9966	.9967	.9968	.9969	.9970

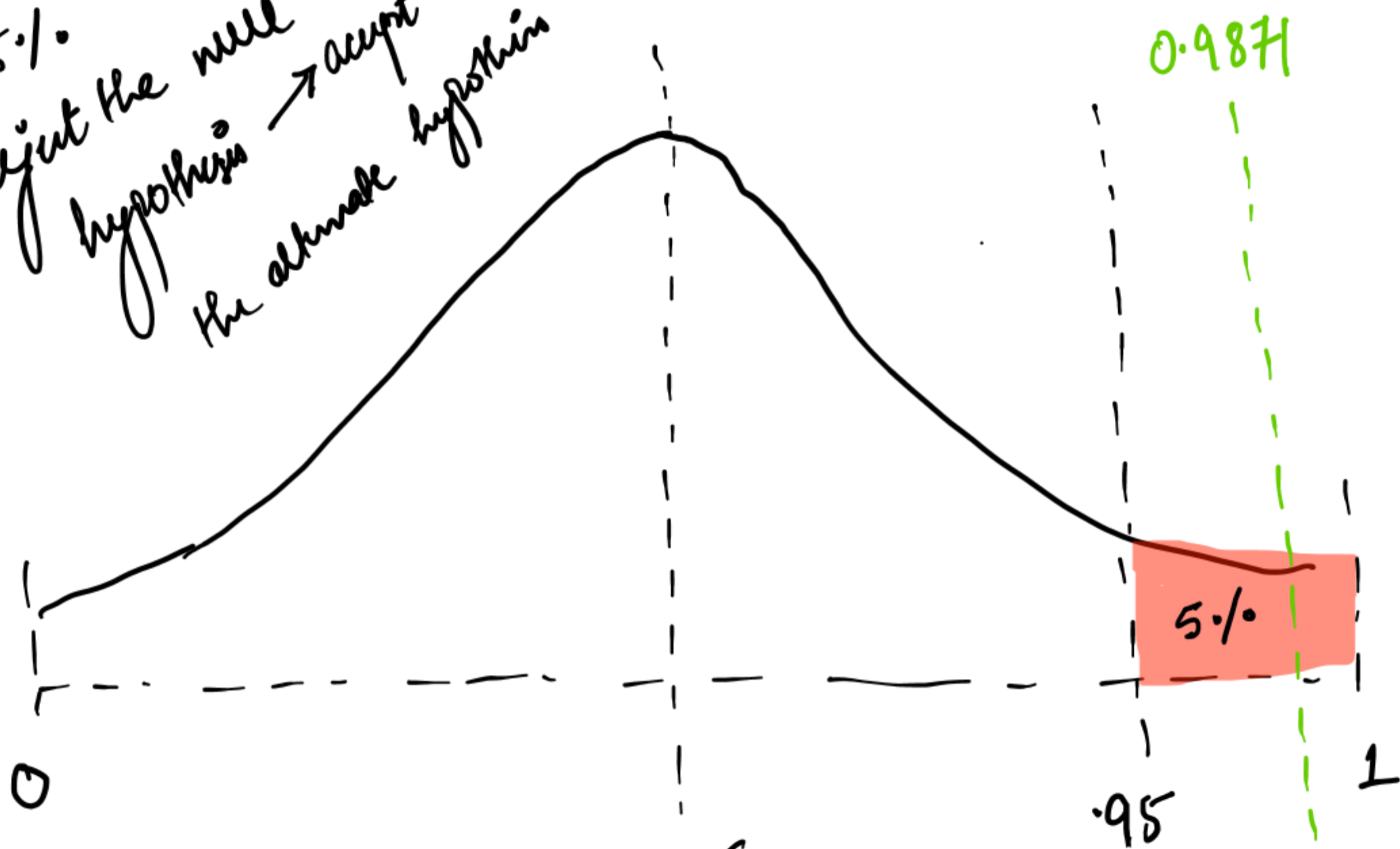
For z-value of 2.256, table entry = 0.9871

p-value $P(\bar{X} \geq z) = \text{unshaded region} = 1 - 0.9871 = 0.01$



= 0.9871

$\alpha = 5\%$
reject the null hypothesis $\rightarrow$ accept the alternate hypothesis



$\rightarrow$ Supermarket should launch 0.5 the loyalty program

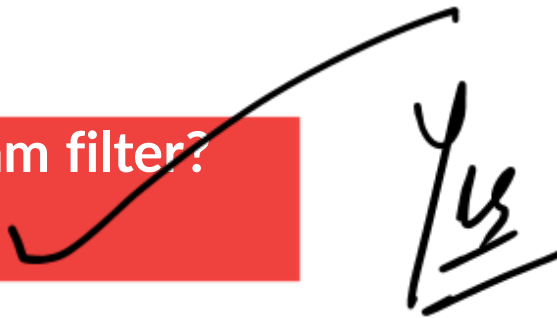
SPAM FILTER

- ABC company is considering buying a commercial filtering software costing \$15,000/annum that the vendor claims will reduce spam to less than 20%.
- The company believes that if the software can reduce spam to less than 20%, then it can help improve employee productivity.
- The company collected a sample of 100 incoming mails with the spam filter and found that 11 of them were spam.
- Assume significance level $\alpha = 0.05$

$$11/100 = 11\%$$

- $\sigma = 0.4$

Should the company buy the spam filter?



① formulating hypothesis :-
 $H_1: \text{Span} < 0.2$ ✓
 $H_0: \text{Span} \geq 0.2$ ✓

② Which test :- left tail test ✓

③ Z-Stat :-

$$\frac{\bar{X} - \mu}{\sigma / \sqrt{n}} = \frac{0.11 - 0.2}{0.4 / \sqrt{100}} = \frac{-0.09}{0.04} = -\frac{9}{4} = \underline{\underline{-2.25}}$$

④ p-value corresponding to Z-value of -2.25

p-value = 0.0122

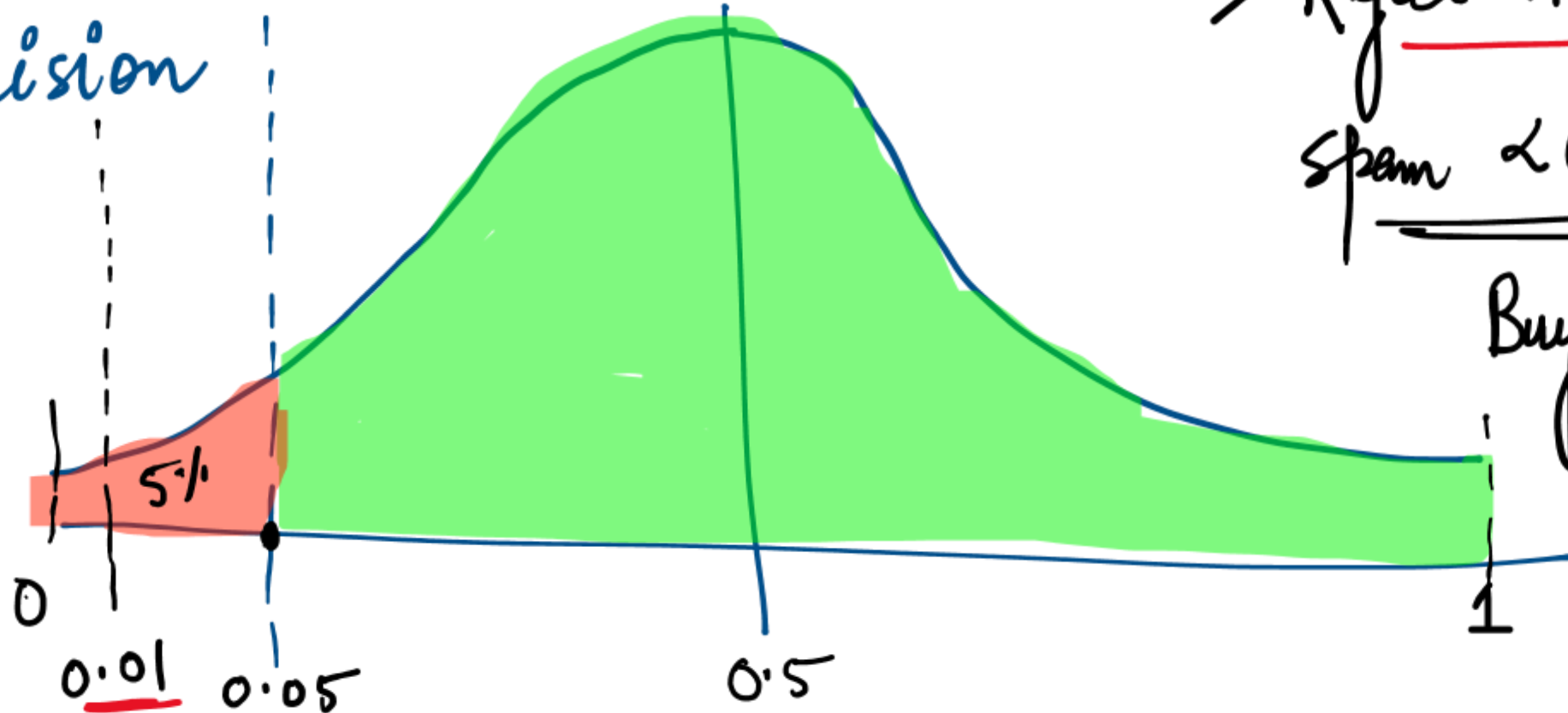
(Z-table)

→ Reject H_0 ✓

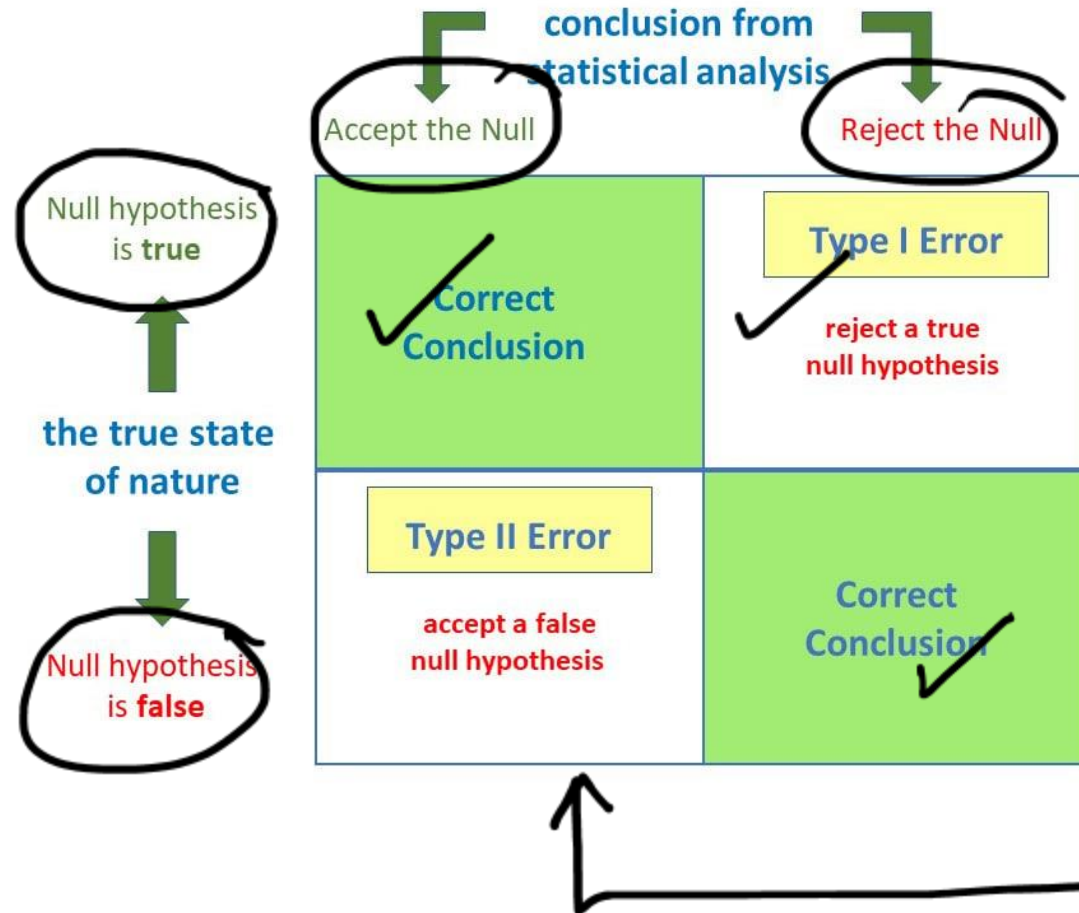
Spam < 0.2

Buy the software

⑤ Decision



→ Errors :-



$N. - T_{true} \rightarrow \text{Reject}$

$N \rightarrow \text{false} \rightarrow \text{Accept}$